

A Multi-Objective Capacitated Vehicle Routing Problem with Optional Service: Problem Definition and Mixed-Integer Formulation

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Abstract

This document defines MO-CVRP-OS, a multi-objective extension of the Capacitated Vehicle Routing Problem (CVRP) in which serving a customer is *optional*, and proposes a mixed-integer linear programming (MILP) formulation for it. The problem is defined directly over the CVRPLIB instances distributed with this repository (the X, XL, AGS and DIMACS families, 222 instances in total), and every modelling assumption is checked against the actual instance data. Five objectives are considered: maximising the number of delivered orders, minimising the number of routes, minimising the largest route diameter, maximising route balance in a max–min (egalitarian) sense, and minimising total travelled distance. The first four are the objectives of interest; the fifth is included because, as shown in Proposition 4.5, the first four are provably blind to the uniform inflation of route lengths. All non-linear elements of the objective vector – a maximum, several minima, and three ratios whose denominators are themselves decision-dependent maxima – are linearised exactly, and the correctness of each linearisation is proved. Modelling choices that the informal specification leaves open are made explicit, and the alternatives are stated and compared.

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1 Introduction

The CVRP asks for a set of minimum-cost vehicle routes, each starting and ending at a single depot, that jointly serve every customer exactly once without violating a common vehicle capacity [10]. It is a single-objective problem: the cost is almost always the total travelled distance, and the service requirement is hard – every customer must be visited.

Both of these features are restrictive in practice. A dispatcher who cannot serve all orders on a given day would rather choose *which* orders to postpone than declare the instance infeasible, and total distance is rarely the only criterion: fleet size, the geographic compactness of each route, and the equitable distribution of work across drivers all matter and conflict with one another. This document formalises one particular combination of these concerns.

Contributions and scope. Section 2 fixes the interpretation of the CVRPLIB instance files shipped in `instances/` and records, as standing assumptions, the properties of that data that the later proofs rely on. Section 3 defines the feasible set at the level of solutions, independently of any formulation. Section 4 defines the five objectives, together with the conventions that make them well defined in degenerate cases. Section 5 recalls the required multi-objective terminology. Section 6 gives the MILP formulation and proves it correct. Section 7 discusses the modelling decisions that were open and the alternatives to the ones taken.

Multi-objective vehicle routing has a substantial literature of its own; see [6] for a survey of the objectives that have been combined with routing and of the solution methods used.

This document is a *formulation* document. It does not propose a solution algorithm, does not report computational experiments, and does not define a benchmarking protocol. The model of Section 6 is a reference formulation: it is exactly solvable only on small instances, whereas the instances in `instances/` have up to 30,000 customers.

2 Instance data and CVRPLIB conventions

2.1 The instance files

The repository contains 222 instances in the TSPLIB-inspired CVRPLIB format, in four families: X (100 instances, [11]), XL (100 instances, [9]), AGS (10 instances, [2]) and DIMACS (12 instances contributed to the 12th DIMACS Implementation Challenge, [3]). Every file declares `TYPE : CVRP`.

A file with `DIMENSION :  $n + 1$` describes $n + 1$ locations numbered $1, \dots, n + 1$. The `DEPOT_SECTION` of every instance in the repository contains the single entry 1, so location 1 is the depot and locations $2, \dots, n + 1$ are the customers. Throughout this document the depot is re-indexed to 0 and customer p of the file becomes customer $p - 1$ of the model, so that

$$V = \{0\} \cup N, \quad N = \{1, 2, \dots, n\}, \quad n = \text{DIMENSION} - 1. \quad (1)$$

The `CAPACITY` field gives the common vehicle capacity $Q \in \mathbb{Z}_{>0}$, and the `DEMAND_SECTION` gives the demands $q_i \in \mathbb{Z}_{\geq 0}$, with $q_0 = 0$ for the depot in all 222 instances.

2.2 Distances

Two `EDGE_WEIGHT_TYPE` values occur, and they must be handled differently.

EUC_2D (210 instances). The `NODE_COORD_SECTION` gives coordinates (a_i, b_i) and the distance is the Euclidean distance *rounded to the nearest integer*, following the TSPLIB95 convention:

$$d_{ij} = \text{nint}\left(\sqrt{(a_i - a_j)^2 + (b_i - b_j)^2}\right), \quad \text{nint}(t) = \lfloor t + \tfrac{1}{2} \rfloor. \quad (2)$$

The instances of the literature that deviate from this convention by using unrounded distances – CMT4, CMT5, tai385 and Golden9–Golden20, as listed in [3] – are not part of this repository, so (2) applies uniformly to all 210 EUC_2D instances here.

EXPLICIT with EDGE_WEIGHT_FORMAT : LOWER_ROW (12 instances). The six Loggi and six ORTEC instances give an explicit integer distance matrix, listed row by row below the diagonal, i.e. row p contains $d_{p,1}, \dots, d_{p,p-1}$. The LOWER_ROW format encodes a symmetric matrix. These are *real road distances*, and they are not metric: sampling 2×10^5 distinct triples per instance finds triangle inequality violations $d_{ij} > d_{ik} + d_{kj}$ in all twelve, in 0.51% to 1.66% of the sampled triples for the six Loggi instances and in 0.011% to 0.054% for the six ORTEC instances. These instances also carry a `NODE_COORD_SECTION`, which is informative only: the explicit matrix takes precedence.

Accordingly, the model below never assumes the triangle inequality, and never uses a valid inequality that would require it.

2.3 Fleet size

No instance in the repository declares a fleet size: the only keys that occur anywhere are NAME, COMMENT, TYPE, DIMENSION, EDGE_WEIGHT_TYPE, EDGE_WEIGHT_FORMAT, NODE_COORD_TYPE and CAPACITY. In particular the `-k` suffix of an instance name is *descriptive, not prescriptive*: it records the number of routes of a reference solution at generation time. This is directly confirmed by the best-known-solution files distributed alongside the instances: of the 112 of them whose name carries a `-k` suffix, 22 use strictly *more* routes than the name advertises and none uses fewer – for instance X-n101-k25.sol contains 26 routes.

The fleet is therefore modelled as homogeneous and unlimited, which is the standard CVRPLIB reading, and the number of vehicles enters the model as an *objective* (Section 4) rather than as a constraint.

2.4 Standing assumptions

The following properties were verified by parsing all 222 instance files; they are used in the proofs of Section 6 and are collected here so that the dependency is explicit.

Assumption 2.1 (Distance matrix). $d : V \times V \rightarrow \mathbb{Z}_{\geq 0}$ satisfies $d_{ii} = 0$ and $d_{ij} = d_{ji}$. The triangle inequality is *not* assumed. Distinct locations may be at distance zero: this occurs for customer pairs in the ten AGS instances, which are the only ones containing duplicate coordinates, and in ten of the twelve EXPLICIT instances, which contain up to four zero-distance customer pairs each.

Assumption 2.2 (Demands). $q_0 = 0$ and $1 \leq q_i \leq Q$ for every customer $i \in N$. Over the 222 instances the customer demands range in $[1, 100]$ and the capacities in $[3, 10,064]$; no instance contains a customer with zero demand, and no instance contains a customer whose demand exceeds the vehicle capacity.

Assumption 2.3 (Depot distances). $d_{0i} \geq 1$ for every customer $i \in N$. No customer of any instance in the repository – EUC_2D or EXPLICIT – lies at distance zero from the depot.

Assumption 2.4 (Orders). Each customer $i \in N$ carries a number of *orders* $o_i \in \mathbb{Z}_{\geq 1}$. CVRPLIB associates exactly one demand quantity with each customer, so the default reading used throughout is $o_i = 1$ for all i , in which case “number of delivered orders” and “number of served customers” coincide. The parameter is kept explicit because the formulation is unchanged for arbitrary positive integer o_i (see Section 7.3).

Assumptions 2.2 and 2.3 are not cosmetic. Assumption 2.2 is what makes the single-commodity flow connectivity constraints of Section 6 valid (Proposition 6.3), and Assumption 2.3 is what makes the route-balance objective well defined without an artificial safeguard (Lemma 4.2). Assumption 2.2 also guarantees that the trivial single-customer route is feasible for every customer, so that the “maximum service” extreme of the Pareto front is attained with all customers served.

3 The problem

3.1 Routes and solutions

Let $G = (V, A)$ be the complete digraph on the node set (1), with arc set $A = \{(i, j) \in V \times V : i \neq j\}$ and arc lengths d_{ij} .

Definition 3.1 (Route). A *route* is a sequence $r = (0, i_1, i_2, \dots, i_m, 0)$ with $m \geq 1$ and $i_1, \dots, i_m$ pairwise distinct customers of N . Its *customer set* is $C(r) = \{i_1, \dots, i_m\}$, and its *length*, *load*, *order count* and *diameter* are

$$L(r) = d_{0i_1} + \sum_{h=1}^{m-1} d_{i_h i_{h+1}} + d_{i_m 0}, \quad (3)$$

$$W(r) = \sum_{i \in C(r)} q_i, \quad (4)$$

$$O(r) = \sum_{i \in C(r)} o_i, \quad (5)$$

$$D(r) = \max_{i, j \in C(r)} d_{ij} = \max(\{0\} \cup \{d_{ij} : i, j \in C(r), i \neq j\}). \quad (6)$$

A route is *capacity feasible* if $W(r) \leq Q$.

Two conventions in Definition 3.1 deserve comment. First, a route is required to be non-empty ($m \geq 1$); an “empty route” is not a route but the absence of one, which is what makes the route-counting objective f_2 below meaningful. Second, the diameter (6) is taken over the customers of the route *only*: the depot is excluded, so a single-customer route has diameter 0. This is the definition adopted throughout; the depot-inclusive and depot-radius alternatives are discussed in Section 7.1.

Definition 3.2 (Solution). A *solution* is a finite set $\mathcal{R} = \{r_1, \dots, r_{|\mathcal{R}|}\}$ of capacity-feasible routes whose customer sets are pairwise disjoint, i.e. $C(r) \cap C(r') = \emptyset$ for all $r \neq r' \in \mathcal{R}$. Its *served set* and *unserved set* are

$$\mathcal{S}(\mathcal{R}) = \bigcup_{r \in \mathcal{R}} C(r), \quad N \setminus \mathcal{S}(\mathcal{R}). \quad (7)$$

The feasible set of MO-CVRP-OS is the set $\mathcal{F}$ of all solutions, including the empty solution $\mathcal{R} = \emptyset$.

Three points distinguish $\mathcal{F}$ from the feasible set of the CVRP.

- (i) *Service is optional.* Definition 3.2 requires the customer sets to be disjoint but does not require them to cover N . Customers left uncovered are simply not served, and no penalty is charged for them: service enters the model through the objective f_1 instead. This places MO-CVRP-OS in the family of routing problems with profits [1], of which the prize-collecting and team-orienteeing problems are the best known members.
- (ii) *No customer is visited twice.* Disjointness of the customer sets together with the distinctness requirement inside a route means that a served customer is visited exactly once, as in the CVRP. Split delivery is not allowed.
- (iii) *The fleet is unlimited.* Definition 3.2 places no bound on $|\mathcal{R}|$; by disjointness and non-emptiness, $|\mathcal{R}| \leq n$ always.

Remark 3.3 (Routes versus route sets). Two distinct routes may have the same customer set but different visiting sequences, and Definition 3.1 treats them as different. This matters: of the five objectives defined next, only $L(\cdot)$ – hence f_4 and f_5 – is sensitive to the visiting order. Proposition 4.5 makes the consequences precise.

4 The objectives

Fix a solution $\mathcal{R} \in \mathcal{F}$. When $\mathcal{R} \neq \emptyset$, write

$$L^{\max} = \max_{r \in \mathcal{R}} L(r), \quad W^{\max} = \max_{r \in \mathcal{R}} W(r), \quad O^{\max} = \max_{r \in \mathcal{R}} O(r), \quad (8)$$

and likewise $L^{\min}, W^{\min}, O^{\min}$ for the corresponding minima.

4.1 Definition

Definition 4.1 (Objective vector). The five objectives of MO-CVRP-OS are

$$f_1(\mathcal{R}) = \sum_{i \in \mathcal{S}(\mathcal{R})} o_i \quad \longrightarrow \quad \max \quad (\text{delivered orders}), \quad (9)$$

$$f_2(\mathcal{R}) = |\mathcal{R}| \quad \longrightarrow \quad \min \quad (\text{routes used}), \quad (10)$$

$$f_3(\mathcal{R}) = \max_{r \in \mathcal{R}} D(r) \quad \longrightarrow \quad \min \quad (\text{largest route diameter}), \quad (11)$$

$$f_4(\mathcal{R}) = \min_{r \in \mathcal{R}} \beta(r) \longrightarrow \max \quad (\text{route balance}), \quad (12)$$

$$f_5(\mathcal{R}) = \sum_{r \in \mathcal{R}} L(r) \longrightarrow \min \quad (\text{total distance}), \quad (13)$$

where the *balance* of a route $r \in \mathcal{R}$ is

$$\beta(r) = \min \left\{ \frac{L(r)}{L^{\max}}, \frac{W(r)/Q}{W^{\max}/Q}, \frac{O(r)}{O^{\max}} \right\} = \min \left\{ \frac{L(r)}{L^{\max}}, \frac{W(r)}{W^{\max}}, \frac{O(r)}{O^{\max}} \right\}. \quad (14)$$

For the empty solution, $f_1(\emptyset) = f_2(\emptyset) = f_3(\emptyset) = f_5(\emptyset) = 0$ and $f_4(\emptyset) = 1$ by convention.

The three components of (14) are exactly the three normalisations required: the route length divided by the longest route length, the capacity utilisation $W(r)/Q$ divided by the largest capacity utilisation among the routes – in which the common factor $1/Q$ cancels, so that the component is simply the load ratio – and the number of delivered orders divided by the largest number of orders on any route. Each component lies in $[0, 1]$ and equals 1 precisely for a route attaining the corresponding maximum.

The combination rule in (14) is the *egalitarian* one: the balance of a route is its worst normalised component, and the balance of the solution is the balance of its worst route. Since a minimum of minima is a minimum over the union,

$$f_4(\mathcal{R}) = \min \left\{ \frac{L^{\min}}{L^{\max}}, \frac{W^{\min}}{W^{\max}}, \frac{O^{\min}}{O^{\max}} \right\}, \quad (15)$$

so f_4 is the worst of the three *spread ratios* of the solution. This identity is what makes the linearisation of Section 6.8 compact: a single scalar variable suffices, instead of one ratio variable per route. Weighted and per-route composite alternatives are discussed in Section 7.2.

4.2 Well-definedness and elementary properties

Lemma 4.2 (The denominators are positive). *For every non-empty solution $\mathcal{R} \in \mathcal{F}$: $L^{\max} \geq L^{\min} \geq 2$, $W^{\max} \geq W^{\min} \geq 1$ and $O^{\max} \geq O^{\min} \geq 1$. Consequently (14) and (15) are well defined, with no division by zero and no need for a safeguard convention.*

Proof. Let $r = (0, i_1, \dots, i_m, 0) \in \mathcal{R}$ with $m \geq 1$. All terms of (3) are non-negative, and the first and last are $d_{0i_1} \geq 1$ and $d_{i_m 0} \geq 1$ by Assumption 2.3; if $m = 1$ these are the same two terms of $L(r) = 2d_{0i_1} \geq 2$, and if $m \geq 2$ they are distinct terms, so $L(r) \geq 2$ in both cases. Similarly $W(r) \geq q_{i_1} \geq 1$ by Assumption 2.2 and $O(r) \geq o_{i_1} \geq 1$ by Assumption 2.4. $\square$

Lemma 4.3 (Range and tightness of f_4). *For every $\mathcal{R} \in \mathcal{F}$, $f_4(\mathcal{R}) \in (0, 1]$. Moreover $f_4(\mathcal{R}) = 1$ if and only if all routes of $\mathcal{R}$ have the same length, the same load and the same order count; in particular $f_4(\mathcal{R}) = 1$ whenever $|\mathcal{R}| \leq 1$.*

Proof. The case $\mathcal{R} = \emptyset$ holds by convention. Otherwise each component of (15) is a ratio of two positive numbers (Lemma 4.2) in which the numerator is a minimum and the denominator the maximum of the same finite family, hence lies in $(0, 1]$; the minimum of three such numbers lies in $(0, 1]$. Equality with 1 holds iff each of the three ratios equals 1, i.e. iff $L^{\min} = L^{\max}$, $W^{\min} = W^{\max}$ and $O^{\min} = O^{\max}$, i.e. iff the three quantities are constant over $\mathcal{R}$. If $|\mathcal{R}| = 1$ this is vacuous. $\square$

Lemma 4.4 (Integrality). *f_1, f_2, f_3 and f_5 take values in $\mathbb{Z}_{\geq 0}$, and f_4 takes values in $\mathbb{Q} \cap (0, 1]$ with denominator at most $\max\{L^{\max}, W^{\max}, O^{\max}\}$.*

Proof. Immediate from Assumptions 2.1, 2.2 and 2.4: all of d, q and o are integer valued, so $L(\cdot), W(\cdot), O(\cdot)$ and $D(\cdot)$ are, and f_4 is a ratio of two such integers. $\square$

4.3 Why a fifth objective

The four objectives $f_1, \dots, f_4$ are the ones of interest: service level, fleet size, geographic compactness and workload fairness. They have, however, a structural blind spot.

Proposition 4.5 (Length degeneracy of $f_1, \dots, f_4$). *Let $\mathcal{R} \in \mathcal{F}$ and let $\mathcal{R}'$ be obtained from $\mathcal{R}$ by permuting the visiting sequence inside one or more routes, leaving all customer sets unchanged. Then*

$$f_1(\mathcal{R}') = f_1(\mathcal{R}), \quad f_2(\mathcal{R}') = f_2(\mathcal{R}), \quad f_3(\mathcal{R}') = f_3(\mathcal{R}), \quad (16)$$

while $f_4(\mathcal{R}')$ and $f_5(\mathcal{R}')$ may differ from $f_4(\mathcal{R})$ and $f_5(\mathcal{R})$. Furthermore, the length ratio in (14) is invariant under a common rescaling of all route lengths, so nothing in $f_1, \dots, f_4$ penalises long routes as such: a solution $\mathcal{R}'$ obtained from $\mathcal{R}$ by lengthening its shorter routes until all route lengths agree satisfies $f_4(\mathcal{R}') \geq f_4(\mathcal{R})$ and $f_5(\mathcal{R}') > f_5(\mathcal{R})$. Consequently, under the objective vector (f_1, f_2, f_3, f_4) alone, a solution may be dominated by one obtained from it purely by lengthening routes.

Proof. f_1 depends on $\mathcal{R}$ only through $\mathcal{S}(\mathcal{R})$, f_2 only through $|\mathcal{R}|$, and f_3 only through the multiset $\{C(r)\}_{r \in \mathcal{R}}$ via (6); none of the three is affected by the visiting order. The same holds for the load and order components of (14), by (4) and (5). Only the component $L(r)/L^{\max}$ sees the sequence, and it is a *ratio*: it rewards routes of *equal* length, not of small length. Example 4.6 exhibits a concrete pair $\mathcal{R}, \mathcal{R}'$ with $f_4(\mathcal{R}') = 1 > f_4(\mathcal{R})$, $f_5(\mathcal{R}') > f_5(\mathcal{R})$ and all other objectives equal. $\square$

Example 4.6 (Levelling down). Take $n = 6$, $Q = 15$, $q_i = 5$ and $o_i = 1$ for all i , with EUC_2D coordinates

$$\text{depot } 0 = (0, 0), \quad 1 = (1, 0), \quad 2 = (2, 0), \quad 3 = (3, 0), \quad 4 = (0, 2), \quad 5 = (0, 3), \quad 6 = (0, 4),$$

so that d is integer valued by (2). Consider the two solutions

$$\mathcal{R} = \{(0, 1, 2, 3, 0), (0, 4, 5, 6, 0)\}, \quad \mathcal{R}' = \{(0, 2, 1, 3, 0), (0, 4, 5, 6, 0)\},$$

which serve the same customers with the same two customer sets $\{1, 2, 3\}$ and $\{4, 5, 6\}$, each of load $15 = Q$. Their route lengths are

$$\begin{aligned} L(0, 1, 2, 3, 0) &= 1 + 1 + 1 + 3 = 6, & L(0, 2, 1, 3, 0) &= 2 + 1 + 2 + 3 = 8, \\ L(0, 4, 5, 6, 0) &= 2 + 1 + 1 + 4 = 8. \end{aligned}$$

Both solutions have $f_1 = 6$, $f_2 = 2$ and $f_3 = \max\{d_{13}, d_{46}\} = \max\{2, 2\} = 2$, and both have all loads and order counts equal, so the load and order components of (15) equal 1 in both. Hence

$$f_4(\mathcal{R}) = \frac{6}{8} = 0.75, \quad f_5(\mathcal{R}) = 14, \quad f_4(\mathcal{R}') = \frac{8}{8} = 1, \quad f_5(\mathcal{R}') = 16.$$

Under (f_1, f_2, f_3, f_4) , $\mathcal{R}'$ dominates $\mathcal{R}$: perfect balance has been achieved by driving two extra kilometres for no reason. Under $(f_1, \dots, f_5)$ the two are incomparable, as they should be, and the trade-off is made explicit to the decision maker.

This phenomenon – an equity criterion improved by making the best-off worse rather than the worst-off better – is well documented in the literature on workload equity in vehicle routing [7], and is the reason why the total distance f_5 is carried as a fifth objective rather than dropped. Section 7.4 records the alternatives (a lexicographic tie-break, an ε -constraint on f_5 , or restricting routes to be sequence-optimal), and how to recover the pure four-objective problem.

Remark 4.7 (Trivial efficient points). The empty solution attains the optimum of f_2, f_3, f_4 and f_5 simultaneously and is therefore always efficient, at the price of the worst possible f_1 . More generally, every single-route solution has $f_4 = 1$ by Lemma 4.3. These points are legitimate but uninformative; if they are to be excluded, the natural device is a minimum service level $f_1 \geq \rho \sum_{i \in N} o_i$ imposed as a constraint (Section 7.8), not a modification of the objectives.

5 Multi-objective terminology

To keep a single convention, collect the objectives of Definition 4.1 into the *minimisation form*

$$F(\mathcal{R}) = (-f_1(\mathcal{R}), f_2(\mathcal{R}), f_3(\mathcal{R}), -f_4(\mathcal{R}), f_5(\mathcal{R})) \in \mathbb{R}^5, \quad (17)$$

so that MO-CVRP-OS reads $\min_{\mathcal{R} \in \mathcal{F}} F(\mathcal{R})$ in the sense of vector optimisation [4].

Definition 5.1 (Dominance and efficiency). For $\mathcal{R}, \mathcal{R}' \in \mathcal{F}$, $\mathcal{R}$ *dominates* $\mathcal{R}'$, written $\mathcal{R} \prec \mathcal{R}'$, if $F_t(\mathcal{R}) \leq F_t(\mathcal{R}')$ for all $t \in \{1, \dots, 5\}$ with at least one strict inequality. A solution $\mathcal{R}^* \in \mathcal{F}$ is *efficient* (Pareto optimal) if no $\mathcal{R} \in \mathcal{F}$ dominates it; the set of efficient solutions is the *efficient set* $\mathcal{F}_E$, and its image $F(\mathcal{F}_E)$ is the *non-dominated set* or *Pareto front*. An efficient solution is *supported* if it minimises $\sum_t w_t F_t(\cdot)$ over $\mathcal{F}$ for some weight vector $w \in \mathbb{R}_{>0}^5$, and *unsupported* otherwise.

Definition 5.2 (Ideal and nadir points). The *ideal point* is $F_t^{\text{id}} = \min_{\mathcal{R} \in \mathcal{F}} F_t(\mathcal{R})$ and the *nadir point* is $F_t^{\text{nad}} = \max_{\mathcal{R} \in \mathcal{F}_E} F_t(\mathcal{R})$, for $t = 1, \dots, 5$.

Two consequences of Section 4 are worth stating. First, $\mathcal{F}$ is finite, so $\mathcal{F}_E \neq \emptyset$ and the Pareto front is finite; by Lemma 4.4 it is a finite subset of $\mathbb{Z}^3 \times \mathbb{Q} \times \mathbb{Z}$ up to the sign convention. Second, $\mathcal{F}$ is a set of combinatorial objects and not a convex set, so the Pareto front is in general non-convex and contains unsupported points; weighted-sum scalarisation cannot enumerate it, whereas the ε -constraint method can [4]. Section 7.7 shows that the ε -constraint treatment of f_4 is particularly convenient here, because bounding f_4 from below by a *constant* removes the only non-linearity of the model.

Part of the ideal point is available in closed form. Since the empty solution is feasible, f_2, f_3, f_5 attain the value 0 and f_4 attains the value 1, so

$$F^{\text{id}} = \left(-\sum_{i \in N} o_i, 0, 0, -1, 0 \right), \quad (18)$$

where the first component uses that serving every customer is always possible (put each customer on its own route; feasible by Assumption 2.2). The ideal point is thus never attained by a single solution as soon as $n \geq 1$, which is the usual situation in multi-objective optimisation and confirms that the five criteria genuinely conflict.

6 Mixed-integer linear formulation

6.1 Overview and design decisions

Three of the five objectives are *route-level*: f_2 counts routes, f_3 is a maximum over routes, and f_4 compares routes to one another. A model must therefore be able to name an individual route, which rules out the classic two-index CVRP formulation in which routes are recovered only implicitly from the arc-incidence vector. The formulation below is consequently *vehicle indexed* (three index): a set K of interchangeable vehicles is introduced, and route k is the tour performed by vehicle k . Section 7.6 discusses the two alternatives – a two-index model with auxiliary route-identification machinery, and a set-partitioning model in which routes are explicit columns.

Vehicle indexing costs two things, and both are addressed below: symmetry among the interchangeable vehicles (Section 6.6) and a factor $|K|$ in the model size (Section 6.12).

6.2 Sets, parameters and constants

Symbol	Meaning
$N = \{1, \dots, n\}$	customers
$V = \{0\} \cup N$	nodes; 0 is the depot
$A = \{(i, j) \in V^2 : i \neq j\}$	arcs
$K = \{1, \dots, \bar{K}\}$	vehicles (interchangeable, identical)
$d_{ij} \in \mathbb{Z}_{\geq 0}$	distance, symmetric, $d_{ii} = 0$ (Assumption 2.1)
$q_i \in \mathbb{Z}_{\geq 1}, i \in N$	demand, $q_i \leq Q$ (Assumption 2.2)
$o_i \in \mathbb{Z}_{\geq 1}, i \in N$	orders, default 1 (Assumption 2.4)
$Q \in \mathbb{Z}_{> 0}$	vehicle capacity
$\bar{K}$	fleet-size bound; default $\bar{K} = n$
$\bar{L}$	upper bound on a route length
$\bar{O} = \sum_{i \in N} o_i$	upper bound on a route order count
$P \in \mathbb{Z}_{\geq 1}$	resolution parameter of the balance linearisation
$\Gamma = 2^P$	number of grid steps, see Section 6.8

Fleet-size bound. By Definition 3.2 every route is non-empty and customer sets are disjoint, so $|\mathcal{R}| \leq n$ and $\bar{K} = n$ excludes no solution. Any smaller $\bar{K}$ is a genuine restriction of $\mathcal{F}$, and one that is not innocuous here: the solutions that minimise f_3 are typically those that split customers into many small routes – with $\bar{K} = n$, putting every customer on its own route gives $f_3 = 0$ – so truncating $\bar{K}$ cuts away exactly the part of the Pareto front where f_3 is best. When a smaller $\bar{K}$ is used for tractability this must be reported as part of the problem instance.

Length bound. $\bar{L}$ must exceed the length of any single route. A route uses at most $n + 1$ arcs, so

$$\bar{L} = (n + 1) d^{\max}, \quad d^{\max} = \max_{(i,j) \in A} d_{ij} \quad (19)$$

is valid. A tighter valid bound, useful because $\bar{L}$ appears in big- M coefficients, is obtained from the capacity: a route carries load at most Q , so it visits at most $m^{\max} = \max\{m : \text{the } m \text{ smallest demands sum to at most } Q\}$ customers, and $\bar{L} = (m^{\max} + 1) d^{\max}$ is valid as well. Both bounds are computed from the instance data and require no assumption on d .

6.3 Decision variables

$$x_{ij}^k \in \{0, 1\}, \quad (i, j) \in A, \quad k \in K \quad \text{vehicle } k \text{ traverses } (i, j), \quad (20)$$

$$y_i^k \in \{0, 1\}, \quad i \in N, \quad k \in K \quad \text{vehicle } k \text{ serves customer } i, \quad (21)$$

$$z_i \in \{0, 1\}, \quad i \in N \quad \text{customer } i \text{ is served at all}, \quad (22)$$

$$u^k \in \{0, 1\}, \quad k \in K \quad \text{vehicle } k \text{ performs a route}, \quad (23)$$

$$g_{ij} \geq 0, \quad (i, j) \in A \quad \text{load carried on arc } (i, j), \quad (24)$$

$$D^k \geq 0, \quad k \in K \quad \text{diameter of route } k, \quad (25)$$

$$L^k, W^k, O^k \geq 0, \quad k \in K \quad \text{length, load, order count of route } k, \quad (26)$$

$$D^{\max}, L^{\max}, W^{\max}, O^{\max} \geq 0, \quad \text{the corresponding maxima}, \quad (27)$$

$$\lambda \in [0, 1], \quad \text{route balance}, \quad (28)$$

$$b_p \in \{0, 1\}, \quad p \in \{0, \dots, P\} \quad \text{binary expansion of } \lambda, \quad (29)$$

$$t_p^L, t_p^W, t_p^O \geq 0, \quad p \in \{0, \dots, P\} \quad \text{linearised products}. \quad (30)$$

It is convenient to abbreviate the *aggregated* arc variables

$$x_{ij} = \sum_{k \in K} x_{ij}^k, \quad (i, j) \in A, \quad (31)$$

which may be substituted out or kept as defined variables.

6.4 Routing, assignment and capacity

$$\sum_{k \in K} y_i^k = z_i, \quad i \in N, \quad (32)$$

$$\sum_{j \in V \setminus \{i\}} x_{ij}^k = y_i^k, \quad i \in N, \quad k \in K, \quad (33)$$

$$\sum_{j \in V \setminus \{i\}} x_{ji}^k = y_i^k, \quad i \in N, \quad k \in K, \quad (34)$$

$$\sum_{j \in N} x_{0j}^k = \sum_{j \in N} x_{j0}^k = u^k, \quad k \in K, \quad (35)$$

$$\sum_{i \in N} q_i y_i^k \leq Q u^k, \quad k \in K, \quad (36)$$

$$u^k \leq \sum_{i \in N} y_i^k, \quad k \in K. \quad (37)$$

Constraint (32) is the optional-service constraint: a customer is assigned to at most one vehicle, and z_i records whether it is assigned at all. Replacing “ $= z_i$ ” by “ $= 1$ ” recovers the mandatory-service CVRP. Constraints (33)–(34) are the degree constraints, written so that a customer has in-degree and out-degree 1 in the sub-graph of vehicle k exactly when k serves it. Equation (35) lets each used vehicle leave and re-enter the depot exactly once, which forbids multiple trips per vehicle. Equation (36) is the capacity constraint and simultaneously forces $y_i^k \equiv 0$ when $u^k = 0$, because $q_i \geq 1$ for all i (Assumption 2.2); the explicit linking inequalities $y_i^k \leq u^k$ are therefore redundant, though they strengthen the linear relaxation and may be added. Equation (37) forbids a “used” vehicle with no customers, so that $\sum_k u^k$ counts routes in the sense of Definition 3.1.

6.5 Connectivity

Constraints (33)–(35) force the arc set of each vehicle to be a disjoint union of circuits, but do not yet force those circuits to pass through the depot. Connectivity is imposed by a single-commodity flow of Gavish–Graves type [5] on the aggregated arcs (31), where g_{ij} is interpreted as the load still on board when arc (i, j) is traversed:

$$\sum_{j \in V \setminus \{i\}} g_{ji} - \sum_{j \in V \setminus \{i\}} g_{ij} = q_i z_i, \quad i \in N, \quad (38)$$

$$q_j x_{0j} \leq g_{0j} \leq Q x_{0j}, \quad j \in N, \quad (39)$$

$$q_j x_{ij} \leq g_{ij} \leq (Q - q_i) x_{ij}, \quad i, j \in N, \quad i \neq j, \quad (40)$$

$$g_{i0} = 0, \quad i \in N. \quad (41)$$

Equation (38) states that each served customer absorbs its demand from the flow, and each unserved customer absorbs nothing. Equation (39) bounds the load leaving the depot by the capacity, Equation (40) states that after visiting i the vehicle carries at most $Q - q_i$ and, if it goes on to j , at least q_j , and Equation (41) empties the vehicle on its return leg. The lower bounds in (39)–(40) are valid and tighten the linear relaxation; they are not needed for correctness.

6.6 Symmetry breaking

The vehicles are interchangeable, so every solution has up to $\bar{K}/(\bar{K} - |\mathcal{R}|)!$ encodings. The following two families of constraints select the canonical one in which routes are ordered by increasing lowest-index customer:

$$u^k \geq u^{k+1}, \quad k \in \{1, \dots, \bar{K} - 1\}, \quad (42)$$

$$y_i^k \leq \sum_{j \in N, j < i} y_j^{k-1}, \quad i \in N, k \in K \setminus \{1\}. \quad (43)$$

Lemma 6.1. *Equations (42) and (43) cut off no solution of $\mathcal{F}$ with at most $\bar{K}$ routes: every such solution admits an encoding satisfying them.*

Proof. Given $\mathcal{R}$ with $|\mathcal{R}| = \kappa \leq \bar{K}$, order its routes as $r_1, \dots, r_\kappa$ by increasing $\mu(r) = \min C(r)$; the μ -values are distinct because the customer sets are disjoint and non-empty. Assign r_k to vehicle k for $k \leq \kappa$ and set $u^k = 1$ iff $k \leq \kappa$, which satisfies (42). For (43), let $k \geq 2$ and $i \in N$ with $y_i^k = 1$. Then $k \leq \kappa$ and $i \in C(r_k)$, so $i \geq \mu(r_k) > \mu(r_{k-1})$, and the customer $j = \mu(r_{k-1}) < i$ satisfies $y_j^{k-1} = 1$, so the right-hand side is at least 1. If $y_i^k = 0$ the constraint is trivially satisfied because its right-hand side is non-negative. $\square$

6.7 Route attributes and the diameter

$$L^k = \sum_{(i,j) \in A} d_{ij} x_{ij}^k, \quad k \in K, \quad (44)$$

$$W^k = \sum_{i \in N} q_i y_i^k, \quad k \in K, \quad (45)$$

$$O^k = \sum_{i \in N} o_i y_i^k, \quad k \in K, \quad (46)$$

$$D^k \geq d_{ij} (y_i^k + y_j^k - 1), \quad i, j \in N, i < j, k \in K, \quad (47)$$

$$D^{\max} \geq D^k, \quad k \in K. \quad (48)$$

Equation (47) is the standard product linearisation for the term $d_{ij} y_i^k y_j^k$: it is active only when both customers are on route k , in which case it reads $D^k \geq d_{ij}$, and is implied by $D^k \geq 0$ otherwise since $d_{ij} \geq 0$. Only unordered pairs are needed because d is symmetric (Assumption 2.1). Note that (47) involves the assignment variables only, which is Proposition 4.5 in constraint form.

6.8 Route balance

The maxima are bounded from below, one-sidedly:

$$L^{\max} \geq L^k, \quad W^{\max} \geq W^k, \quad O^{\max} \geq O^k, \quad k \in K, \quad (49)$$

$$L^{\max} \leq \bar{L}, \quad W^{\max} \leq Q, \quad O^{\max} \leq \bar{O}. \quad (50)$$

The upper bounds (50) are valid because every individual route satisfies $L^k \leq \bar{L}$, $W^k \leq Q$ and $O^k \leq \bar{O}$; they are stated explicitly because the product linearisation of Lemma 6.6 requires a finite upper bound on each factor. The balance variable λ is then bounded above by each of the three normalised components of (14), for used routes only. Written directly, this is

$$\lambda \cdot L^{\max} \leq L^k + \bar{L} (1 - u^k), \quad k \in K, \quad (51)$$

$$\lambda \cdot W^{\max} \leq W^k + Q (1 - u^k), \quad k \in K, \quad (52)$$

$$\lambda \cdot O^{\max} \leq O^k + \bar{O}(1 - u^k), \quad k \in K. \quad (53)$$

The big- M coefficients are the smallest valid ones: the left-hand sides are bounded by $L^{\max} \leq \bar{L}$, $W^{\max} \leq Q$ and $O^{\max} \leq \bar{O}$ respectively, since $\lambda \leq 1$. An unused vehicle ($u^k = 0$) is thereby excluded from the minimum, as Equation (12) requires.

Constraints (51)–(53) are the only non-linear part of the model: each contains the product of the continuous variable λ with a continuous variable. They are linearised exactly by restricting λ to a uniform grid of $\Gamma + 1 = 2^P + 1$ points in $[0, 1]$ and expanding the grid index in binary:

$$\lambda = \frac{1}{\Gamma} \sum_{p=0}^P 2^p b_p, \quad \sum_{p=0}^P 2^p b_p \leq \Gamma, \quad (54)$$

together with, for each $p \in \{0, \dots, P\}$, the McCormick identities for the products $t_p^L = b_p L^{\max}$, $t_p^W = b_p W^{\max}$ and $t_p^O = b_p O^{\max}$:

$$t_p^L \leq \bar{L} b_p, \quad t_p^L \leq L^{\max}, \quad t_p^L \geq L^{\max} - \bar{L}(1 - b_p), \quad t_p^L \geq 0, \quad (55)$$

and the analogous groups for t_p^W (with $W^{\max}$, Q) and t_p^O (with $O^{\max}$, $\bar{O}$). Substituting $\lambda L^{\max} = \Gamma^{-1} \sum_p 2^p t_p^L$ and its analogues, Equations (51) to (53) become the linear constraints

$$\frac{1}{\Gamma} \sum_{p=0}^P 2^p t_p^L \leq L^k + \bar{L}(1 - u^k), \quad k \in K, \quad (56)$$

$$\frac{1}{\Gamma} \sum_{p=0}^P 2^p t_p^W \leq W^k + Q(1 - u^k), \quad k \in K, \quad (57)$$

$$\frac{1}{\Gamma} \sum_{p=0}^P 2^p t_p^O \leq O^k + \bar{O}(1 - u^k), \quad k \in K. \quad (58)$$

6.9 The model

Definition 6.2 (MO-CVRP-OS MILP).

$$\text{maximise} \quad f_1 = \sum_{i \in N} o_i z_i, \quad (59)$$

$$\text{minimise} \quad f_2 = \sum_{k \in K} u^k, \quad (60)$$

$$\text{minimise} \quad f_3 = D^{\max}, \quad (61)$$

$$\text{maximise} \quad f_4 = \lambda, \quad (62)$$

$$\text{minimise} \quad f_5 = \sum_{k \in K} L^k = \sum_{(i,j) \in A} d_{ij} x_{ij}, \quad (63)$$

subject to

- (32)–(37) assignment, degree, depot and capacity constraints,
- (38)–(41) connectivity,
- (42)–(43) symmetry breaking,
- (44)–(48) route attributes and diameter,
- (49)–(58) route balance,

and to the variable restrictions

$$x, y, z, u, b \text{ binary}, \quad g, D, L, W, O, D^{\max}, L^{\max}, W^{\max}, O^{\max}, t \geq 0, \quad \lambda \in [0, 1]. \quad (64)$$

6.10 Correctness

Proposition 6.3 (The feasible sets correspond). *Let $\bar{K} \geq 1$.*

- (a) *Every (x, y, z, u, g) satisfying (32)–(41) encodes a solution $\mathcal{R} \in \mathcal{F}$ with $|\mathcal{R}| = \sum_k u^k \leq \bar{K}$: for each k with $u^k = 1$ the arcs $\{(i, j) : x_{ij}^k = 1\}$ form a single circuit through the depot, whose customer set is $\{i : y_i^k = 1\}$ and whose load is at most Q ; for each k with $u^k = 0$ that arc set is empty.*
- (b) *Conversely, every $\mathcal{R} \in \mathcal{F}$ with $|\mathcal{R}| \leq \bar{K}$ admits an encoding satisfying (32)–(41) and the symmetry-breaking constraints (42)–(43).*

Proof. (a) Fix k . If $u^k = 0$ then (36) gives $\sum_i q_i y_i^k \leq 0$, hence $y^k \equiv 0$ by Assumption 2.2, hence $x^k \equiv 0$ by (33) and (35). If $u^k = 1$, then by (33)–(35) every node of $\{0\} \cup C_k$, where $C_k = \{i : y_i^k = 1\}$, has out-degree and in-degree exactly 1 in the arc set of vehicle k , and every other node has degree 0; such an arc set is a disjoint union of circuits covering $\{0\} \cup C_k$, and $C_k \neq \emptyset$ by (37).

It remains to exclude circuits avoiding the depot. Summing (32) over k shows that in the aggregated digraph $\hat{G} = (V, \{(i, j) : x_{ij} \geq 1\})$ every customer i has out-degree and in-degree exactly $z_i \in \{0, 1\}$; in particular $x_{ij} \in \{0, 1\}$ for all arcs, and the arcs of the different vehicles are pairwise disjoint. Suppose some vehicle's arc set contained a circuit Z with $V(Z) \subseteq N$. Every $i \in V(Z)$ has $z_i = 1$ and its unique incoming and unique outgoing arc in $\hat{G}$ both belong to Z ; therefore every arc of A with exactly one endpoint in $V(Z)$ has $x_{ij} = 0$, and by (39)–(40) carries $g_{ij} = 0$. Summing (38) over $i \in V(Z)$, the flow on the arcs internal to Z appears once positively and once negatively and cancels, and the arcs leaving $V(Z)$ carry no flow, so the left-hand side vanishes, while the right-hand side equals $\sum_{i \in V(Z)} q_i \geq |V(Z)| \geq 1 > 0$ by Assumption 2.2. This contradiction shows that each vehicle's arc set is a single circuit through the depot, i.e. a route in the sense of Definition 3.1, with customer set C_k . The customer sets are disjoint by (32) and the load bound is (36).

(b) Order the routes of $\mathcal{R}$ as in the proof of Lemma 6.1 and assign route r_k to vehicle k . Set $x_{ij}^k = 1$ for the arcs of r_k , $y_i^k = 1$ for $i \in C(r_k)$, $z_i = 1$ for $i \in \mathcal{S}(\mathcal{R})$, $u^k = 1$ for $k \leq |\mathcal{R}|$, and all remaining variables of these families to 0. Constraints (32)–(37) hold by construction and capacity feasibility. For the flow, let g_{ij} be the load still on board on arc (i, j) of the route containing it, i.e. for $r_k = (0, i_1, \dots, i_m, 0)$ set $g_{i_h i_{h+1}} = \sum_{l > h} q_{i_l}$ for $h = 0, \dots, m$ with the conventions $i_0 = i_{m+1} = 0$, and $g_{ij} = 0$ on all other arcs. Then (38) holds at every customer of a route because consecutive values differ by exactly q_{i_h} , and at every unserved customer because all incident g vanish; (39) holds because $g_{0i_1} = W(r_k) \leq Q$ and $g_{0i_1} \geq q_{i_1}$; (40) holds because $g_{i_h i_{h+1}} = W(r_k) - \sum_{l \leq h} q_{i_l} \leq Q - q_{i_h}$ and $g_{i_h i_{h+1}} \geq q_{i_{h+1}}$ for $h < m$; and (41) holds because $g_{i_m 0} = 0$. The symmetry-breaking constraints hold by Lemma 6.1. $\square$

Remark 6.4 (Where Assumption 2.2 is used). The proof of Proposition 6.3(a) uses $q_i \geq 1$ twice: to deduce $y^k \equiv 0$ from $u^k = 0$, and – essentially – to derive the contradiction $0 = \sum_{i \in V(Z)} q_i > 0$. If an instance admitted a customer with zero demand, a depot-free circuit consisting entirely of such customers would satisfy (38)–(41) and the formulation would be wrong. No instance in this repository has a zero-demand customer, but a formulation intended for arbitrary data should either add the usual ε -demand perturbation, or replace the flow system by generalised subtour elimination constraints (Section 7.5).

Proposition 6.5 (The auxiliary variables report the intended values). *Let $(x, y, z, u, g, \dots)$ be feasible for Definition 6.2 and let $\mathcal{R}$ be the solution it encodes (Proposition 6.3). Then*

$$\sum_{i \in N} o_i z_i = f_1(\mathcal{R}), \quad \sum_{k \in K} u^k = f_2(\mathcal{R}), \quad \sum_k L^k = f_5(\mathcal{R}), \quad (65)$$

identically, while

$$D^{\max} \geq f_3(\mathcal{R}) \quad \text{and} \quad \lambda \leq \lfloor f_4(\mathcal{R}) \rfloor_{\Gamma}, \quad (66)$$

where $\lfloor \cdot \rfloor_{\Gamma}$ rounds down to the grid $\{0, 1/\Gamma, \dots, 1\}$. Both bounds in (66) are attained by some feasible completion of the same (x, y, z, u, g) . Consequently, for any scalarisation that is non-decreasing in $D^{\max}$ and non-increasing in λ – weighted sums with non-negative weights, ε -constraints, lexicographic orderings, and any combination of these – an optimal solution of Definition 6.2 reports $D^{\max} = f_3(\mathcal{R})$ and $\lambda = \lfloor f_4(\mathcal{R}) \rfloor_{\Gamma}$.

Proof. The identities (65) are immediate from (32), (37) and (44), the last because the arc sets of distinct vehicles are disjoint.

For $D^{\max}$: by (47), $D^k \geq d_{ij}$ for every pair $i, j \in C_k$, so $D^k \geq D(r_k)$ and $D^{\max} \geq \max_k D(r_k) = f_3(\mathcal{R})$ by (48); setting $D^k = D(r_k)$ and $D^{\max} = f_3(\mathcal{R})$ is feasible, since (47) is then satisfied with equality for a maximising pair and slack elsewhere, and (48) holds by definition of a maximum. As $D^{\max}$ occurs in no other constraint and is minimised, it equals $f_3(\mathcal{R})$ at any optimum.

For λ : by (49), $L^{\max} \geq \max_k L^k = \max_{r \in \mathcal{R}} L(r)$ in the notation of (8), and similarly for $W^{\max}, O^{\max}$. For any used route k , (51) gives $\lambda \leq L^k / L^{\max} \leq L^k / \max_{r \in \mathcal{R}} L(r)$ and analogously for the other two families, so $\lambda \leq \beta(r_k)$ for every k with $u^k = 1$, whence $\lambda \leq f_4(\mathcal{R})$; since (54) confines λ to the grid, $\lambda \leq \lfloor f_4(\mathcal{R}) \rfloor_{\Gamma}$. Conversely, setting $L^{\max} = \max_r L(r)$, $W^{\max} = \max_r W(r)$, $O^{\max} = \max_r O(r)$, $\lambda = \lfloor f_4(\mathcal{R}) \rfloor_{\Gamma}$ with the corresponding b , and $t_p^L = b_p L^{\max}$, $t_p^W = b_p W^{\max}$, $t_p^O = b_p O^{\max}$ is feasible: (55) and its analogues hold with equality by Lemma 6.6 below, and (56)–(58) reduce to $\lambda \max_r L(r) \leq L^k$, $\lambda \max_r W(r) \leq W^k$ and $\lambda \max_r O(r) \leq O^k$ for used k , which hold because $\lambda \leq f_4(\mathcal{R}) \leq \beta(r_k)$. Unused k are covered by the big- M terms, which are valid by $\lambda L^{\max} \leq \bar{L}$, $\lambda W^{\max} \leq Q$, $\lambda O^{\max} \leq \bar{O}$. Since $L^{\max}, W^{\max}, O^{\max}$ occur only in (49) and in (51)–(53), where decreasing them relaxes the constraint, and λ is maximised, the stated values are attained. $\square$

Lemma 6.6 (Exactness of the product linearisation). *Let $b \in \{0, 1\}$, let v be a variable with $0 \leq v \leq \bar{v}$, and let t satisfy $t \leq \bar{v}b$, $t \leq v$, $t \geq v - \bar{v}(1 - b)$ and $t \geq 0$. Then $t = bv$. Consequently, with (54) and (55), $\Gamma^{-1} \sum_p 2^p t_p^L = \lambda L^{\max}$ exactly, and analogously for W and O ; the constraints (56)–(58) are therefore equivalent to (51)–(53) restricted to λ on the grid.*

Proof. If $b = 0$ the first and fourth inequalities give $t = 0 = bv$. If $b = 1$ the second and third give $v \leq t \leq v$, so $t = v = bv$. Summing, $\Gamma^{-1} \sum_p 2^p t_p^L = \Gamma^{-1} \sum_p 2^p b_p L^{\max} = \lambda L^{\max}$ by (54). $\square$

Corollary 6.7 (Resolution and exact recovery). *Let $M = \max\{\bar{L}, Q, \bar{O}\}$, let $\mathcal{R}$ be the solution encoded by an optimal solution of Definition 6.2 and let λ be the reported balance.*

- (a) $0 \leq f_4(\mathcal{R}) - \lambda < 2^{-P}$. In particular the model is exact for every value of f_4 that is a dyadic rational with denominator dividing Γ , and approximate otherwise.
- (b) Let $\mathbb{Q}_M = \{a/c : c \in \{1, \dots, M\}, a \in \{0, \dots, c\}\}$, so that $f_4(\mathcal{R}) \in \mathbb{Q}_M$ by Lemma 4.4. If $P \geq 2 \log_2 M$, then $[\lambda, \lambda + 2^{-P})$ contains exactly one element of $\mathbb{Q}_M$, namely $f_4(\mathcal{R})$; the exact value is therefore recovered from λ in post-processing, for instance by a Stern–Brocot search.

Proof. (a) The grid $\{0, 1/\Gamma, \dots, 1\}$ has mesh $1/\Gamma = 2^{-P}$, and Proposition 6.5 gives $\lambda = \lfloor f_4(\mathcal{R}) \rfloor_{\Gamma}$. (b) Let $a/c \neq a'/c'$ be elements of $\mathbb{Q}_M$. Then $|a/c - a'/c'| = |ac' - a'c|/(cc') \geq 1/(cc') \geq 1/M^2 \geq 2^{-P}$, the last step by $P \geq 2 \log_2 M$. Two distinct elements of $\mathbb{Q}_M$ inside $[\lambda, \lambda + 2^{-P})$ would differ by strictly less than 2^{-P} , a contradiction, and $f_4(\mathcal{R})$ lies in that interval by (a). $\square$

Remark 6.8 (Exactness without discretisation). Corollary 6.7 is a statement about a genuine approximation: the model of Definition 6.2 is an exact MILP for f_4 restricted to a grid, not

for f_4 itself. Three exact alternatives exist and are compared in Section 7.7: fixing $\lambda = \theta$ as a constant, which makes (51)–(53) linear and is what an ε -constraint scheme does anyway; bisection on θ ; and keeping (51)–(53) as bilinear constraints and handing the model to a global MIQCP solver. A fourth option keeps the structure of (54)–(58) but replaces the dyadic grid by an arbitrary finite set $\Theta \subseteq [0, 1]$, selected by binaries s_θ with $\sum_{\theta \in \Theta} s_\theta = 1$ and $\lambda = \sum_{\theta \in \Theta} \theta s_\theta$; taking $\Theta = \mathbb{Q}_M$ as in Corollary 6.7 makes the model exact, at the price of $\Theta(M^2)$ binaries instead of $P + 1$.

6.11 Bounds and valid inequalities

Proposition 6.9. *The following are valid for Definition 6.2.*

- (a) Fleet lower bound. $\sum_{k \in K} u^k \geq \frac{1}{Q} \sum_{i \in N} q_i z_i$, and, when z is fixed, the rounded form $\sum_k u^k \geq \lceil \frac{1}{Q} \sum_i q_i z_i \rceil$.
- (b) Diameter–fleet interaction. If $\sum_k u^k \leq \kappa$ and $\sum_i z_i = \sigma \geq 1$, then some used route serves at least $\lceil \sigma/\kappa \rceil$ customers, so

$$D^{\max} \geq \min \left\{ \max_{i,j \in S} d_{ij} : S \subseteq N, |S| = \lceil \sigma/\kappa \rceil \right\}.$$

*In particular $f_3 = 0$ forces every used route to have pairwise zero-distance customers, which by Assumption 2.1 is possible only in the ten AGS instances and the twelve **EXPLICIT** ones, and even there only for very small routes. Without a bound on $\sum_k u^k$ no non-trivial lower bound on $D^{\max}$ exists, since putting every customer on its own route yields $f_3 = 0$. The bound above is stated for insight: evaluating it is itself NP-hard in general.*

- (c) Balance upper bound. $\lambda \leq 1$, attained iff all used routes have equal length, load and order count (Lemma 4.3).
- (d) Distance lower bound. Both of

$$f_5 \geq \left(\min_{(i,j) \in A} d_{ij} \right) \sum_{i \in N} z_i \quad \text{and} \quad f_5 \geq \frac{1}{2} \sum_{i \in N} z_i \left(\min_{j \neq i} d_{ij} + \min_{j \neq i} d_{ji} \right)$$

are valid, and neither uses the triangle inequality.

Proof. (a) Sum (36) over k and use (32). The rounded form follows from integrality of $\sum_k u^k$. (b) If at most κ routes serve σ customers, the pigeonhole principle puts at least $\lceil \sigma/\kappa \rceil$ of them on one route, whose diameter is by (6) at least the smallest diameter of a customer set of that cardinality; the displayed bound then follows from (48). (c) is Lemma 4.3. (d) Each served customer has exactly one outgoing and one incoming arc in $\hat{G}$ by (33)–(34), whose lengths are at least the stated minima; each arc is counted at most twice. $\square$

Proposition 6.10 (Complexity). *MO-CVRP-OS is NP-hard, and deciding whether a feasible solution with $f_1 = \sum_i o_i$, $f_2 \leq \kappa$ and $f_5 \leq B$ exists is NP-complete.*

Proof. Restricting $\mathcal{F}$ to solutions with $\mathcal{S}(\mathcal{R}) = N$ and $|\mathcal{R}| \leq \kappa$, and minimising f_5 , is exactly the CVRP with κ vehicles, which is NP-hard [10]; membership in NP of the decision version is clear since a solution is a polynomial-size certificate whose objectives are computable in polynomial time. $\square$

6.12 Model size

Family	Count	Type
x_{ij}^k	$\bar{K} n(n+1)$	binary
y_i^k	$\bar{K} n$	binary
z_i	n	binary
u^k	$\bar{K}$	binary
b_p	$P+1$	binary
g_{ij}	$n(n+1)$	continuous
D^k, L^k, W^k, O^k	$4\bar{K}$	continuous
$D^{\max}, L^{\max}, W^{\max}, O^{\max}, \lambda$	5	continuous
t_p^L, t_p^W, t_p^O	$3(P+1)$	continuous
(47) (diameter)	$\bar{K} \binom{n}{2}$	
(33)–(34), (43)	$\Theta(\bar{K}n)$	
(38)–(41)	$\Theta(n^2)$	
(49)–(50), (56)–(58)	$6\bar{K} + 3$	
(55) and analogues	$12(P+1)$	

With the default $\bar{K} = n$ the model has $\Theta(n^3)$ binary variables and $\Theta(n^3)$ constraints, dominated by the arc variables and by the diameter constraints (47). The balance machinery, by contrast, costs only $P+1$ additional binaries and $O(\bar{K}+P)$ constraints, which is the payoff of the collapse (15).

This is a reference formulation. Direct solution is realistic only for instances with a few tens of customers, whereas the smallest instance in **instances/** has $n = 100$ and the largest $n = 30,000$. For those, the formulation serves to define the problem unambiguously, to validate heuristics on small sub-instances, and as the starting point for the decompositions of Section 7.6.

7 Modelling decisions and alternatives

The informal specification of MO-CVRP-OS leaves several points open. This section states the decision taken in each case, and the alternatives that were rejected, so that the model can be varied deliberately rather than by accident. Table 1 summarises.

7.1 Route diameter

Equation (6) measures the geographic spread of the *customers* of a route. Three alternatives are natural.

- (a) *Depot-inclusive diameter* $D'(r) = \max_{i,j \in C(r) \cup \{0\}} d_{ij}$, linearised by adding $D^k \geq d_{0i} y_i^k$ for $i \in N$ to (47). It never vanishes on a non-empty route (by Assumption 2.3), so the degenerate “one customer per route gives $f_3 = 0$ ” behaviour of Proposition 6.9(b) disappears and f_3 becomes $\max_r \max\{D(r), \max_{i \in C(r)} d_{0i}\}$. This makes f_3 conflict with f_2 much more strongly, which may be desirable or not.
- (b) *Depot radius* $\rho(r) = \max_{i \in C(r)} d_{0i}$, linearised by $\rho^k \geq d_{0i} y_i^k$. It measures how far a route ventures rather than how spread out it is, and it is cheaper: $O(\bar{K}n)$ constraints instead of $O(\bar{K}n^2)$.
- (c) *Longest leg* $\max_{(i,j) \in r} d_{ij}$, linearised by $D^k \geq d_{ij} x_{ij}^k$. Unlike the other three, this depends on the visiting sequence, so Proposition 4.5 would not apply to it.

Open point	Decision	Main alternatives
Route diameter	customers only, (6)	depot-inclusive; depot radius; longest leg (Section 7.1)
Balance combiner	egalitarian min, (14)	weighted per-route composite; component-wise minima (Section 7.2)
Balance normaliser	ratio to the maximum	range; standard deviation; Gini; Jain (Section 7.2)
Length degeneracy	fifth objective f_5	lexicographic tie-break; ε -constraint; sequence-optimal routes (Section 7.4)
Service objective	$\sum_i o_i z_i, o_i = 1$	delivered demand $\sum_i q_i z_i$; profits (Section 7.3)
Connectivity	single-commodity flow	MTZ; generalised SECs; multi-commodity flow (Section 7.5)
Route identification	vehicle indices	two-index; set partitioning (Section 7.6)
Fleet bound	$\bar{K} = n$	any $\bar{K} < n$, at the cost of part of the front (Section 6.2)
f_4 exactness	grid of mesh 2^{-P}	fixed θ ; bisection; bilinear MIQCP (Section 7.7)

Table 1: Modelling decisions.

The chosen definition (6) has one property worth naming: it depends on the *partition* of the served customers only. Minimising f_3 alone, subject to serving everybody with at most κ routes, is therefore the min-max diameter clustering problem, which is NP-hard; f_3 contributes genuine combinatorial difficulty on top of the routing.

7.2 Combining the balance components

Write $\ell_r = L(r)/L^{\max}$, $v_r = W(r)/W^{\max}$ and $\omega_r = O(r)/O^{\max}$ for the three normalised components of route r , and let $\alpha, \beta, \gamma \geq 0$ with $\alpha + \beta + \gamma = 1$. Three combination rules were considered:

$$f_4^{\text{egal}}(\mathcal{R}) = \min_{r \in \mathcal{R}} \min\{\ell_r, v_r, \omega_r\} \quad (\text{adopted, (14)}), \quad (67)$$

$$f_4^{\text{comp}}(\mathcal{R}) = \min_{r \in \mathcal{R}} (\alpha \ell_r + \beta v_r + \gamma \omega_r) \quad (\text{per-route composite, then min}), \quad (68)$$

$$f_4^{\text{cw}}(\mathcal{R}) = \alpha \min_r \ell_r + \beta \min_r v_r + \gamma \min_r \omega_r \quad (\text{component-wise minima, then combine}). \quad (69)$$

Proposition 7.1. *For every $\mathcal{R} \in \mathcal{F}$ and all admissible weights,*

$$f_4^{\text{egal}}(\mathcal{R}) \leq f_4^{\text{cw}}(\mathcal{R}) \leq f_4^{\text{comp}}(\mathcal{R}), \quad (70)$$

and all three equal 1 simultaneously, precisely under the conditions of Lemma 4.3.

Proof. For the right inequality, $\min_r$ of a sum is at least the sum of the $\min_r$'s of its terms, applied to (68) with non-negative weights. For the left, $\min\{a, b, c\} \leq \alpha a + \beta b + \gamma c$ whenever $\alpha + \beta + \gamma = 1$ and $\alpha, \beta, \gamma \geq 0$; apply this with $a = \min_r \ell_r$ and so on, and note $f_4^{\text{egal}} = \min\{\min_r \ell_r, \min_r v_r, \min_r \omega_r\}$ by (15). All three equal 1 iff each ratio equals 1 for every route, which is Lemma 4.3. $\square$

The adopted rule (67) is thus the most conservative of the three, requires no weights to be chosen, and – by the collapse (15) – costs a single scalar variable λ . Equation (68) is the most

permissive: a route may be poor in one dimension and compensate in another. Implementing (68) requires one ratio variable π_r per route and component, hence $3\bar{K}(P+1)$ binaries instead of $P+1$; (69) requires three, which is still cheap.

Rules that are not ratios to the maximum are also possible: the range $L^{\max} - L^{\min}$ (to be minimised, and linear in the model variables, so it would need no linearisation at all), the standard deviation of the route lengths (convex quadratic), the Gini coefficient (linearisable with $O(\bar{K}^2)$ absolute-value variables), or Jain’s fairness index $(\sum_k L^k)^2 / (|\mathcal{R}| \sum_k (L^k)^2)$ (a ratio of quadratics). The survey of [7] compares such measures for vehicle routing specifically and documents the pathology of Proposition 4.5 – an equity measure improved by making the best-off worse – for min-max and range-based measures alike, which is precisely why f_5 is retained here.

7.3 The service objective

Equation (9) counts orders. With the default $o_i = 1$ of Assumption 2.4 it counts served customers. Two alternatives require no change to the formulation, only to the coefficients of (59): *delivered demand* $\sum_i q_i z_i$, appropriate when the quantity shipped matters more than the number of stops, obtained by setting $o_i = q_i$; and *profit* $\sum_i p_i z_i$ for instance-specific prizes p_i , which is the classical prize-collecting objective [1]. Note that the choice also propagates into the third component of the balance measure (14), which is defined in terms of $O(r) = \sum_{i \in C(r)} o_i$; if orders and balance should be counted differently, two separate parameter vectors must be introduced.

7.4 Handling the length degeneracy

Proposition 4.5 forces a choice. Four responses are available.

- (a) *A fifth objective* (adopted). The trade-off between balance and distance becomes explicit and is left to the decision maker. Cost: a five-dimensional Pareto front, which is harder to compute and to present.
- (b) *A lexicographic tie-break*. Keep the four-objective vector and refine the dominance relation of Definition 5.1 by declaring $\mathcal{R} \prec \mathcal{R}'$ also when $F_{1:4}(\mathcal{R}) = F_{1:4}(\mathcal{R}')$ and $f_5(\mathcal{R}) < f_5(\mathcal{R}')$. The resulting efficient set is a subset of the four-objective one containing, for each attainable (f_1, f_2, f_3, f_4) vector, only the shortest solutions achieving it. This is the least intrusive fix and is recommended when a four-objective front is required for comparability with other work.
- (c) *An ε -constraint on distance*. Keep four objectives and add $f_5 \leq \bar{d}$ for an instance-specific budget, e.g. a multiple of a best-known CVRP solution value. Sound, but the front then depends on an exogenous parameter.
- (d) *Sequence-optimal routes*. Redefine $\mathcal{F}$ to contain only solutions in which each route is a minimum-length Hamiltonian circuit on $C(r) \cup \{0\}$. This removes the degeneracy at the source – Proposition 4.5 becomes vacuous because the sequence is a function of the customer set – and reduces MO-CVRP-OS to a pure partitioning problem. The price is that feasibility becomes NP-hard to verify, so the condition has no compact MILP expression; it is natural only in the set-partitioning model of Section 7.6, where it amounts to restricting the column set Ω to sequence-optimal routes.

Dropping f_5 from Definition 6.2 recovers the pure four-objective problem; nothing else in the formulation changes.

7.5 Connectivity constraints

The single-commodity flow of Section 6.5 was chosen because it is compact, because it is stronger than the Miller–Tucker–Zemlin alternative, and because its validity argument survives optional service. The alternatives:

- (a) *Miller–Tucker–Zemlin* [8], per vehicle: variables s_i^k with $q_i \leq s_i^k \leq Q$ and

$$s_i^k - s_j^k + Q x_{ij}^k \leq Q - q_j, \quad i, j \in N, \ i \neq j, \ k \in K.$$

Compact and simple, but its linear relaxation is dominated by that of (38)–(40), and it needs $\Theta(\bar{K}n^2)$ constraints against $\Theta(n^2)$ for the aggregated flow.

- (b) *Generalised subtour elimination constraints*, in the form adapted to optional service:

$$\sum_{(i,j) \in A, i,j \in S} x_{ij} \leq \sum_{i \in S} z_i - z_h, \quad S \subseteq N, \ |S| \geq 2, \ h \in S.$$

Exponentially many, separated in polynomial time by minimum-cut computations. Exact without Assumption 2.2, which makes it the right choice if zero-demand customers must be supported (Remark 6.4).

- (c) *Directed capacity constraints* $\sum_{(i,j) \in \delta^+(S)} x_{ij} \geq \frac{1}{Q} \sum_{i \in S} q_i z_i$ for $S \subseteq N$. Valid as written; note that the familiar rounded form $\lceil q(S)/Q \rceil$ is *not* directly available here, because the right-hand side depends on the variables z , and rounding is legitimate only after z has been fixed (for example in a branch where the served set is decided).
- (d) *Multi-commodity flow*, one commodity per served customer: the strongest compact relaxation, at $\Theta(n^3)$ variables, which is not attractive at the sizes considered here.

7.6 Alternative formulations

Two-index model. Replacing x_{ij}^k by x_{ij} removes the factor $\bar{K}$ from the model size and the vehicle symmetry with it, and suffices for f_1 , f_5 and connectivity. It does not suffice for f_2 , f_3 or f_4 : the number of routes is recoverable as $\sum_j x_{0j}$, but the diameter and the balance measure need to know *which* customers share a route, which the aggregated arc vector does not express without reintroducing an assignment structure. The undirected variant, with $e_{ij} \in \{0, 1\}$ for $i, j \in N$ and $e_{0j} \in \{0, 1, 2\}$ to allow single-customer routes, has the same limitation.

Set partitioning. Let Ω be the set of all capacity-feasible routes, $a_{ir} = 1$ if route $r \in \Omega$ visits customer i , and $\theta_r \in \{0, 1\}$ select route r . With L_r , W_r , O_r , D_r the (constant) attributes of column r :

$$\sum_{r \in \Omega} a_{ir} \theta_r = z_i \leq 1, \quad i \in N, \quad (71)$$

$$f_1 = \sum_i o_i z_i, \quad f_2 = \sum_{r \in \Omega} \theta_r, \quad f_5 = \sum_{r \in \Omega} L_r \theta_r, \quad (72)$$

$$D^{\max} \geq D_r \theta_r, \quad r \in \Omega, \quad (73)$$

$$L^{\max} \geq L_r \theta_r, \quad \lambda \cdot L^{\max} \leq L_r + \bar{L}(1 - \theta_r), \quad r \in \Omega, \quad (74)$$

and analogously for W and O . Connectivity and capacity are enforced implicitly by the definition of Ω , all four route attributes are constants, and the sequence-optimality of Section 7.4(d) is imposed simply by restricting Ω . The obstacle is that $D^{\max}$, $L^{\max}$, $W^{\max}$, $O^{\max}$ and λ couple the columns, so the linear relaxation is not a pure set-partitioning LP and its pricing problem is

not the usual elementary shortest path problem with resource constraints: the dual information attached to the coupling constraints must be carried into the pricing. A practical route is to fix $(D^{\max}, L^{\max}, W^{\max}, O^{\max}, \lambda)$ in an outer enumeration or bisection – see Section 7.7 – and to solve a standard branch-and-price for each fixed tuple.

7.7 Scalarisation, and exact treatment of f_4

Because $\mathcal{F}$ is combinatorial, the Pareto front has unsupported points and weighted-sum scalarisation cannot enumerate it; the ε -constraint method can [4]. For MO-CVRP-OS this is doubly convenient.

- (a) f_4 as a constraint is linear. Imposing $f_4 \geq \theta$ for a constant $\theta \in [0, 1]$ turns (51)–(53) into

$$\theta L^{\max} \leq L^k + \bar{L}(1 - u^k), \quad \theta W^{\max} \leq W^k + Q(1 - u^k), \quad \theta O^{\max} \leq O^k + \bar{O}(1 - u^k),$$

which are linear. The entire apparatus (54)–(58) can then be deleted, and the model is an exact MILP with no discretisation error whatsoever (Remark 6.8). Since f_4 takes finitely many rational values (Lemma 4.4), bisection on θ down to the separation $1/M^2$ of Corollary 6.7 computes $\max f_4$ exactly in $O(\log M)$ MILP solves, with $M = \max\{\bar{L}, Q, \bar{O}\}$.

- (b) f_3 as a constraint is a preprocessing rule. Imposing $f_3 \leq \varepsilon_3$ is equivalent to forbidding two customers at distance more than ε_3 from sharing a route:

$$y_i^k + y_j^k \leq 1 \quad \text{for all } i < j \text{ with } d_{ij} > \varepsilon_3, \quad k \in K,$$

which also lets the variables D^k and $D^{\max}$ and the constraints (47)–(48) be dropped entirely. This is a far better behaved model than the min-max form and is the recommended way to explore the f_3 dimension.

- (c) *Bilinear form.* Keeping (51)–(53) as written yields a mixed-integer bilinear program (MIQCP) that a global solver can handle directly. This is exact and compact but gives up the MILP machinery; it is mentioned for completeness, not recommended.
- (d) *Dinkelbach.* If only one of the three components of (14) is retained – say the length component, so that $f_4 = L^{\min}/L^{\max}$ – then maximising f_4 is a single-ratio mixed-integer fractional program, solvable exactly by Dinkelbach’s algorithm: iterate $\max\{L^{\min} - \theta L^{\max}\}$ and update $\theta \leftarrow L^{\min}/L^{\max}$ at the incumbent. The sum-of-ratios structure of (68) and (69) does not admit this treatment, which is a further argument for the egalitarian rule (67).

7.8 Optional side constraints

None of the following is part of MO-CVRP-OS as defined, but each is a one-line addition to Definition 6.2 and each is standard in applications.

$$\sum_{i \in N} o_i z_i \geq \rho \sum_{i \in N} o_i, \quad \text{minimum service level } \rho \in [0, 1] \text{ (see Remark 4.7),} \quad (75)$$

$$z_i = 1, \quad i \in M \subseteq N, \quad \text{mandatory customers,} \quad (76)$$

$$\sum_{k \in K} u^k \leq \kappa, \quad \text{fleet cap,} \quad (77)$$

$$L^k \leq \bar{L}^{\text{route}}, \quad k \in K, \quad \text{maximum route length or duration.} \quad (78)$$

The first is the principled way to exclude the trivial efficient points of Remark 4.7; note that it changes the ideal point and therefore any normalisation built on it.

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